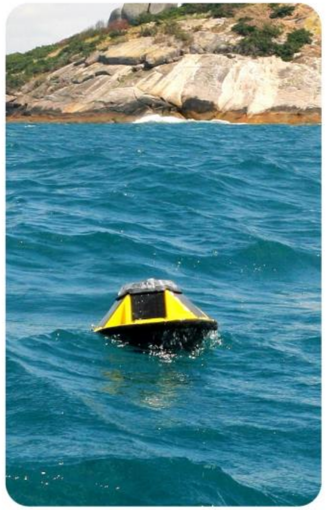
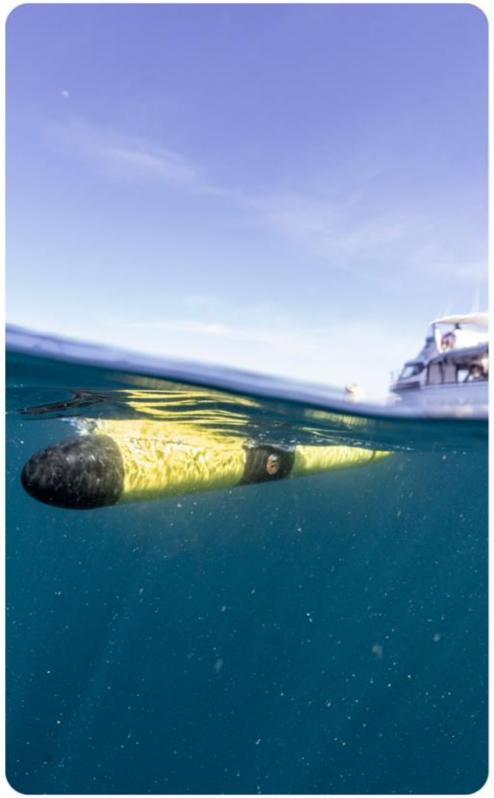
IMOS Integrated Marine Observing System

Biological Ocean Observer





Claire H. Davies[1] and Jason D. Everett[2,3]
[1]CSIRO Environment, Hobart
[2]The University of Queensland
[3]Everdat Solutions Pty Ltd

IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work and observe, and recognise them as Australia's first marine scientists and carers of Sea Country. We pay our respects to Aboriginal and Torres Strait Islander peoples past and present.

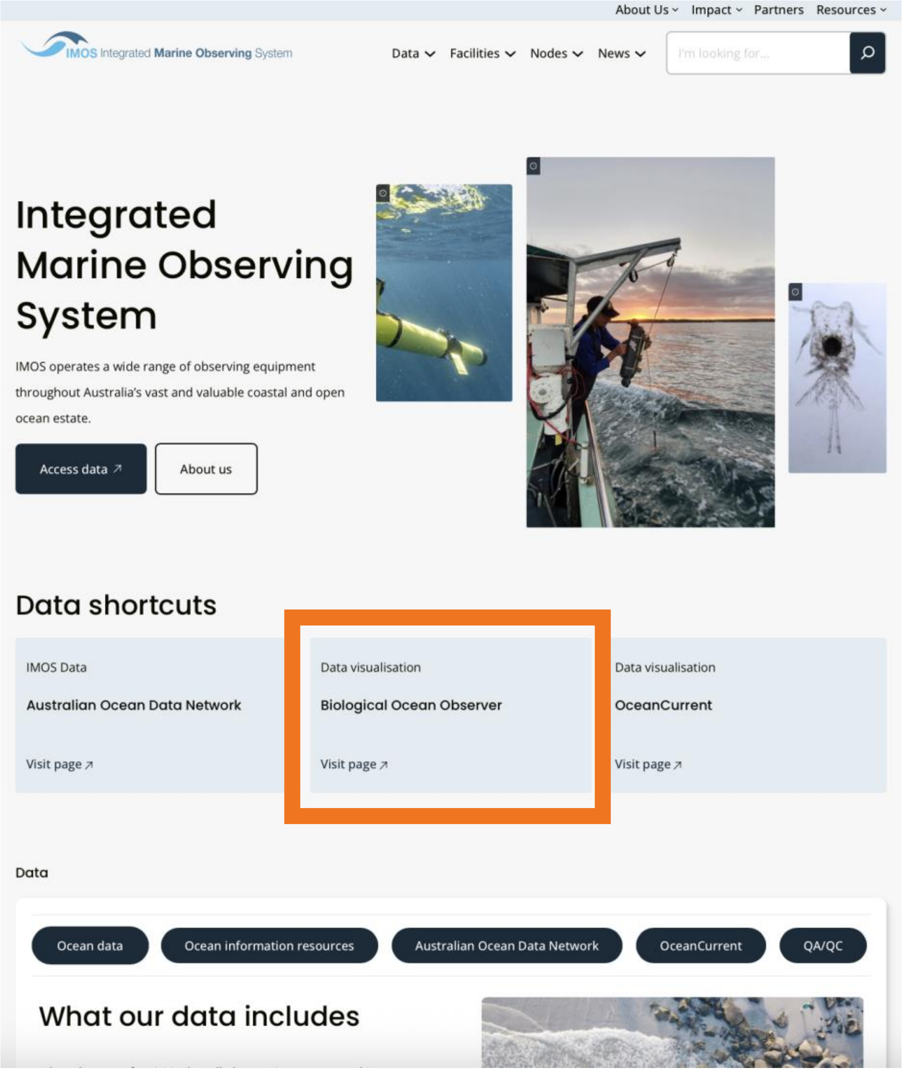
Where, what, why??

shiny.csiro.au/BioOceanObserver

What can I use it for:

- accessing biological data
- visualising biological data
- publication ready plotting
- accessing tutorials

github.com/PlanktonTeam



IMOS Integrated Marine Observing System

IMOS Integrated Marine Observing System Biological Ocean Observer

Home EOVs Microbes Phytoplankton Zooplankton Larval Fish Environmental Data Relationships Information CSIRO

Welcome Sampling Progress Sampling Status Sampling Summary

Funded by:

IMOS Integrated Marine Observing System

ARDC Australian Research Data Commons

The Biological Ocean Observer

The goal of this site is to Integrate, Analyse and Visualise data collected by the Integrated Marine Observing System (IMOS). We aim to enhance the availability and understanding of biological data and make it accessible to broader and non-specialist audiences in order to accelerate the next generation of scientific insights.

This project is entirely open source, as are all the IMOS data underlying it. All the code for this tool are freely available on GitHub. We welcome collaborators and pull requests are gratefully accepted. If you encounter a problem with this website or have any feedback, please log an issue on GitHub or email the IMOS Office.

This tool was originally conceived and developed by Dr Jason Everett (UQ/CSIRO/UNSW) and Claire Davies (CSIRO). Jason is a biological oceanographer and Claire is a plankton ecologist. Both have a strong interest in open data science and encouraging increased data uptake to solve real world problems.

The major categories of data we provide within the app are found across the top bar, and include microbial, phytoplankton, zooplankton, larval fish and environment (chemical) parameters. The snapshot and EOV tab include summary data that may be useful as data overviews for managers and policy makers. Within each tab, the data is often designated by sampling regime, which is generally the National Reference Stations (NRS) or the Continuous Plankton Recorder (CPR). Due to the spatial nature of the CPR data, these data are summarised by Australia's Marine Bioregions. More information about the methods used in this tool, can be found in 'Technical Information' under the 'Information' tab.

Citation

If you use this app in any publication, please cite as:
'Davies, Claire H., Everett, Jason D., Ord, Louise (2022) Integrated Marine Observing System (IMOS) Biological Ocean Observer - Shiny APP. v9.3. CSIRO. Service Collection. http://hdl.handle.net/102.100.100/447365?index=1.

All of the analysis and plotting contained in this application are powered by the *planktonr* package:
Everett Jason D., Davies Claire H. (2022). planktonr: Analysis and visualisation of plankton data. R package version 0.5.6.0000, https://github.com/PlanktonTeam/planktonr.

Acknowledging IMOS Data

This application is developed using IMOS data, and therefore you are also required to clearly acknowledge the source material by including the following statement in any publications:
'Data were sourced from Australia's Integrated Marine Observing System (IMOS) - IMOS is enabled by the National Collaborative Research Infrastructure Strategy

planktonr

planktonr 0.5.6.0000 | Get started | Reference | Articles

3. Phytoplankton

Source: `vignettes/Phytoplankton.Rmd`

Contents

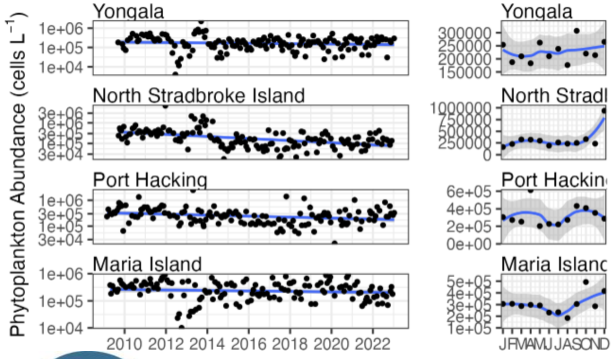
Trend Analysis

Here we plot the abundance of phytoplankton as a tim for the National Reference Stations around Australia. This can give us trends. At this time, early 2023, phytoplankton abundance seem to be generally decreasing along the east coast and this may have implications for zooplankton and higher trophic levels as phytoplankton is thought of as the base of the food web.

```
NRSp <- planktonr::pr_get_Indices("NRS", "P") %>%
  filter(Parameters == "PhytoAbundance_CellsL") %>%
  filter(StationCode %in% c("YON", "NSI", "PHB", "MAI"))

p1 <- planktonr::pr_plot_Trends(NRSp, Trend = "Raw", Survey = "NRS", method
= "lm", trans = "log10")
p2 <- planktonr::pr_plot_Trends(NRSp, Trend = "Month", Survey = "NRS", metho
d = "loess")

p1 + p2 +
  ggplot2::theme(axis.title.y = ggplot2::element_blank()) + # Remove y-title
from 2nd column
  patchwork::plot_layout(widths = c(3, 1), guides = "collect")
```



IMOS Integrated Marine Observing System

planktonr

planktonteam.github.io/planktonr

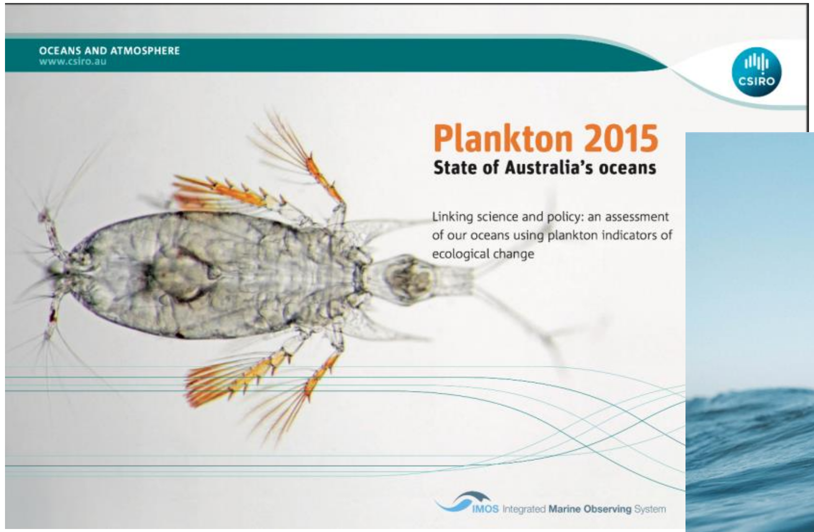
Biological Ocean Observer

shiny.csiro.au/BioOceanObserver

Integrates with R through *planktonr*

https://planktonteam.github.io/planktonr

Uptake & Impact – Policy & Management



NSW GOVERNMENT
Department of Climate Change, Energy, Environment and Water

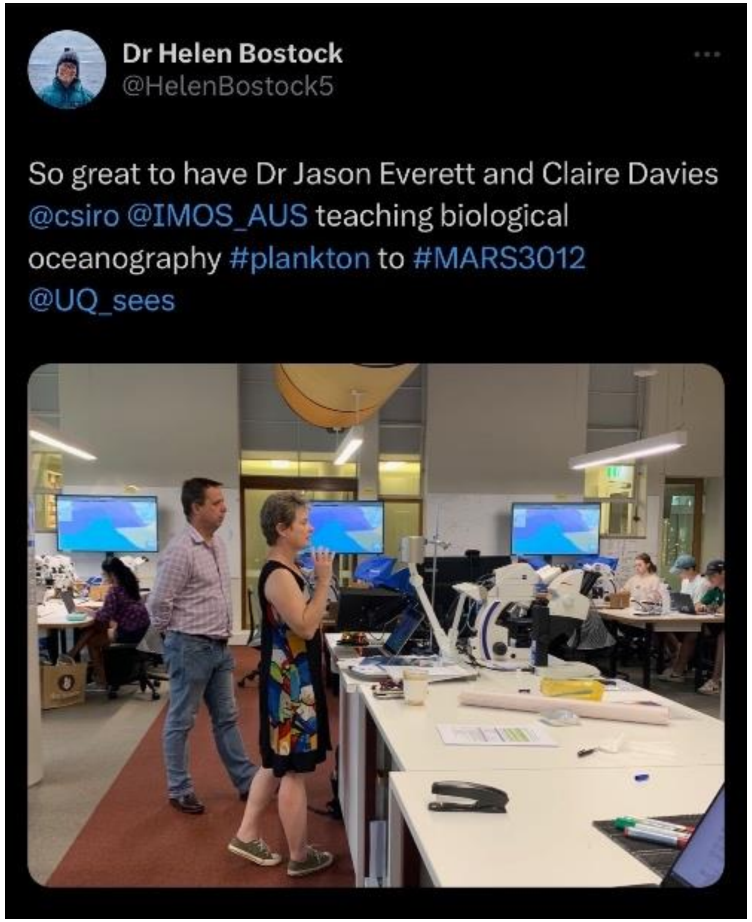
Changing Oceans: Science and Synthesis – A regional analysis of the state of NSW coastal waters 2025.

Uptake & Impact – Teaching & research

- Teaching
  - **UQ: Physical-Biological Oceanography** (MARS3012) to contextualise Moreton Bay field work
  - **UQ: Physical-Biological Oceanography** (MARS3012) State of Australia's Oceans Report
  - **UTas: Oceanographic Methods** (KSA324) to explore how oceanography influences biology

- University Research
  - Josh Esman (UNSW) – Species abundance of copepods and larval fish
  - Mark Koh (UQ) – Temporal & spatial variation of zooplankton

- Industry Research and Engagement
  - Data analytics for regional scale trends in Chl.
  - Demonstrated at the Coastal Conference 2023



Future Development

Collaborating with other IMOS teams who are developing similar tools.

Plenty of synergies

Southern Ocean CPR Data

Industry HAB data

NESP Marine and Coastal Hub (2025).
Making marine environmental data more assessment-ready (NESP 5.9 Project)

IMOS Southern Ocean Time Series

RLS
REEF LIFE SURVEY

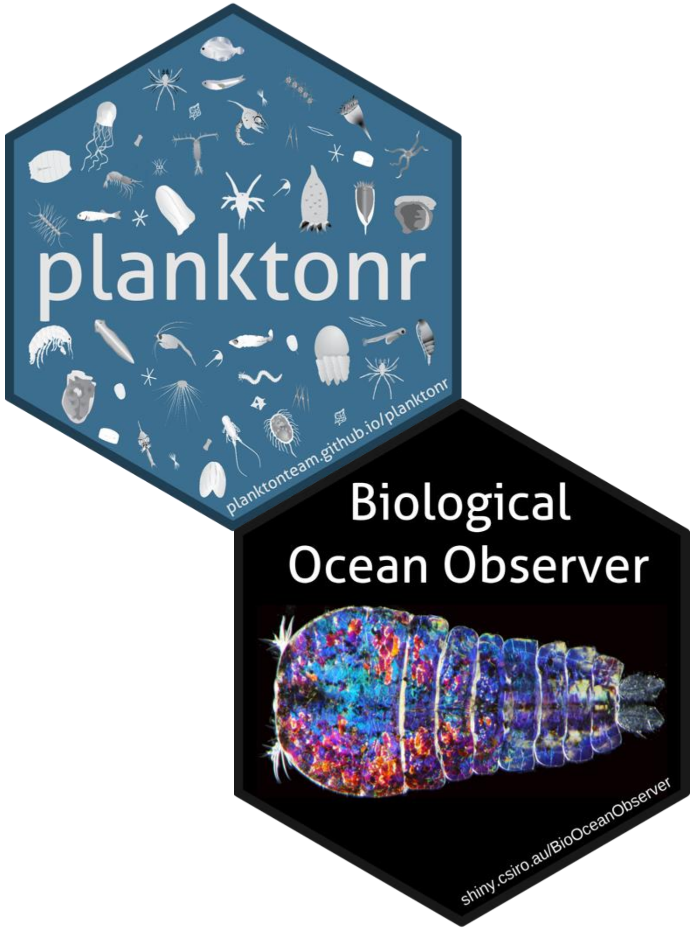
planktonr
planktonteam.github.io/planktonr

remora

Biological Ocean Observer
shiny.csiro.au/BioOceanObserver

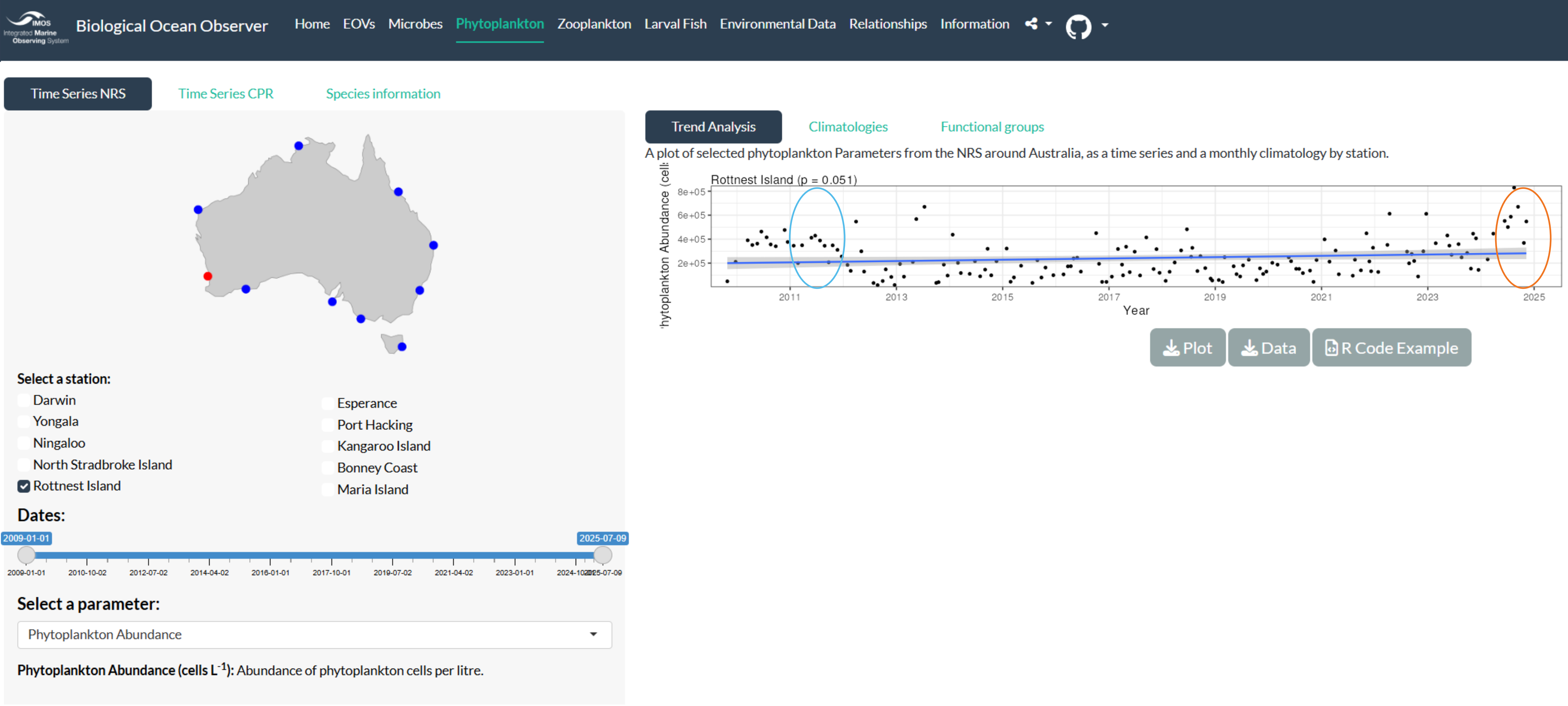
IMOS OceanCurrent
Surface Currents and Temperature

Case Studies



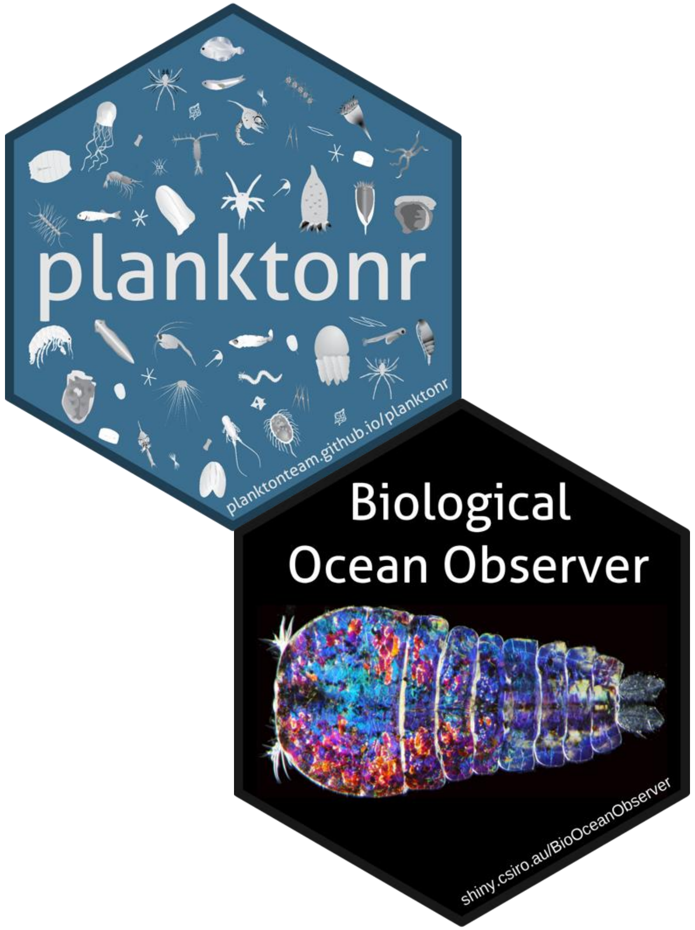
- **WA summer 2024 / 25 heat**
  - **How is the biology affected by long periods of warming**

- South Australia Algal Bloom
  - What is the biological response
  - Other IMOS data sources



IMOS Integrated Marine Observing System

Case Studies



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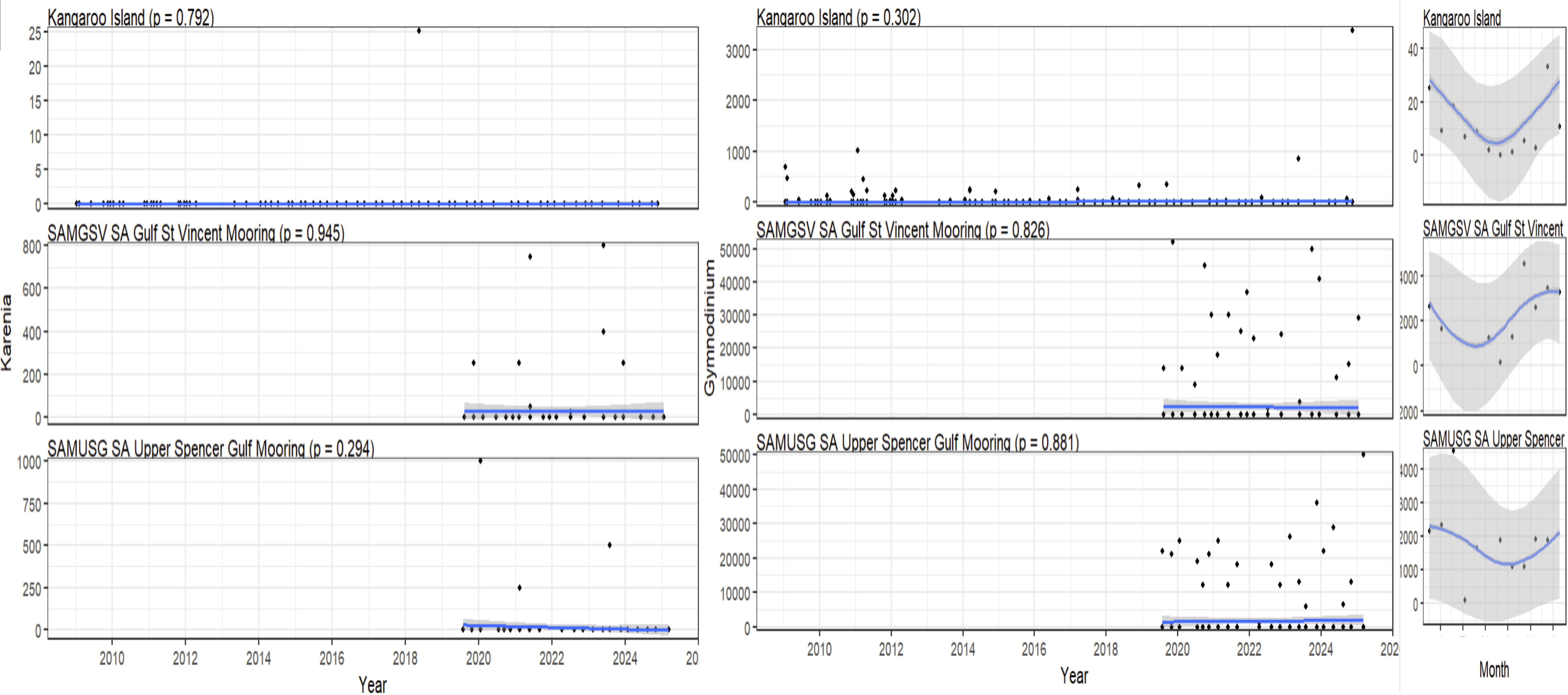


Using planktonr functions to investigate species abundances

IMOS Integrated **Marine Observing** System

Case Studies

- WA summer 2024 / 25 heat
  - How is the biology affected by long periods of warming
- **South Australia Algal Bloom**
  - What is the biological response
  - **Other IMOS data sources**

BIOPLATFORMS AUSTRALIA
DATA PORTAL

Australian Microbiome

planktonr
planktonteam.github.io/planktonr

Biological Ocean Observer
shiny.csiro.au/BioOceanObserver

BIOPLATFORMS AUSTRALIA
DATA PORTAL

claire.davies@csiro.au

methods manual

Filter by amplicon, taxonomy and traits

Amplicon | is | 18Sv4F18Sv4R_v4_eukaryote

Taxonomy

Select Amplicon to filter taxonomy

Sample search | BLAST search | Interactive map search | Interactive graph search

Download OTU and Contextual Data (CSV) | Download Contextual Data only (CSV) | Download BIOM format (Phinch compatible) | Export Data to Galaxy Australia for further analysis | Export Data to Galaxy Australia for Krona Taxonomic Abundance Graph

Found 169 samples

Sample ID	Environment	Latitude	Longitude
139719	Marine	-35.8322	136.4473
139720	Marine	-35.8322	136.4473
139721	Marine	-35.8322	136.4473
139722	Marine	-35.8322	136.4473

Trait | is | ---

Clear

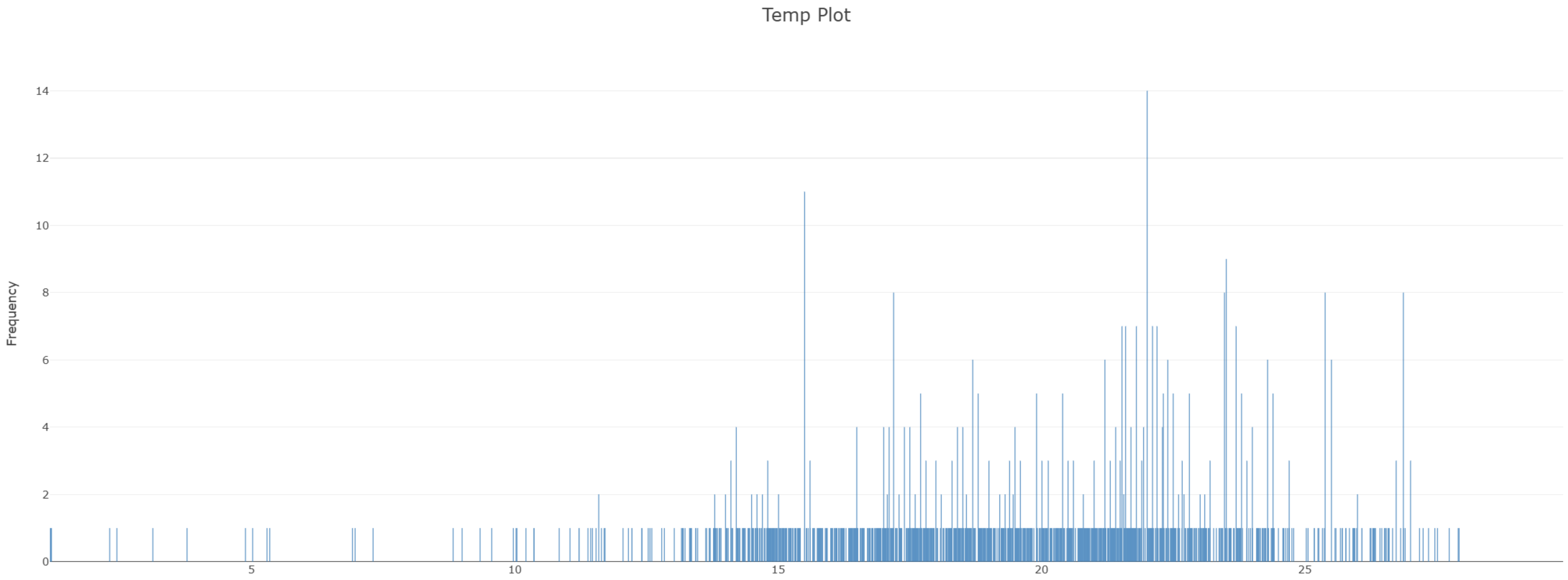
IMOS Integrated Marine Observing System

https://data.bioplatforms.com/bpa/otu

Interactive Graphical Search | Tabbed View | List View | Tutorial

Amplicon | Taxonomy | Traits | Environment | Contextual Filters | Taxonomy vs Environment | Traits vs Environment

Vegetation Type | Env Broad Scale | Env Local Scale | pH | Sample Type | Temp | Nitrate Nitrite | Nitrite | Chlorophyll Ctd | Salinity | Silicate | Latitude | Longitude



All conte

Amplicon <is> 18Sv4F18Sv4R_v4_eukaryote

Taxonomy <is> silva138 SKlearn

Domain <is> d__Eukaryota

Phylum <is> p__Dinoflagellata

Class <is> c__Dinophyceae

Order <is> o__Gymnodiniphycidae

Family <is> f__Kareniaceae

AM Environment <is> Marine

Latitude <between> -39.65194037907667 and -26.505185132543318

Longitude <between> 126.87731444163776 and 159.79204624967213

IMOS Integrated Marine Observing System

BIOPLATFOR
AUSTRALIA
DATA PORTAL

Tutorial Help Australian Microbiome Home amotu@example.org

BLAST Search

The BLAST search tool implements a blastn search over sequences returned by the taxonomy search and contextual filters implemented. For best results, submit a query that spans only the amplicon region, or set `qcov_hsp_perc` to sensible value given your query sequence. We've set the default value to `60`, which will (for query length ~= amplicon length) omit usually unwanted short local alignments.

The download provides: 1) the blast search details, 2) the blast table results, 3) a table with hit sequences, their match details, their locations and sample attributes, and 4) a map showing hit locations and sequence similarities.

Note that the figure is sorted to plot the highest scoring points last, so any symbols occurring at the same location will only be visible as the highest scoring alignment value.

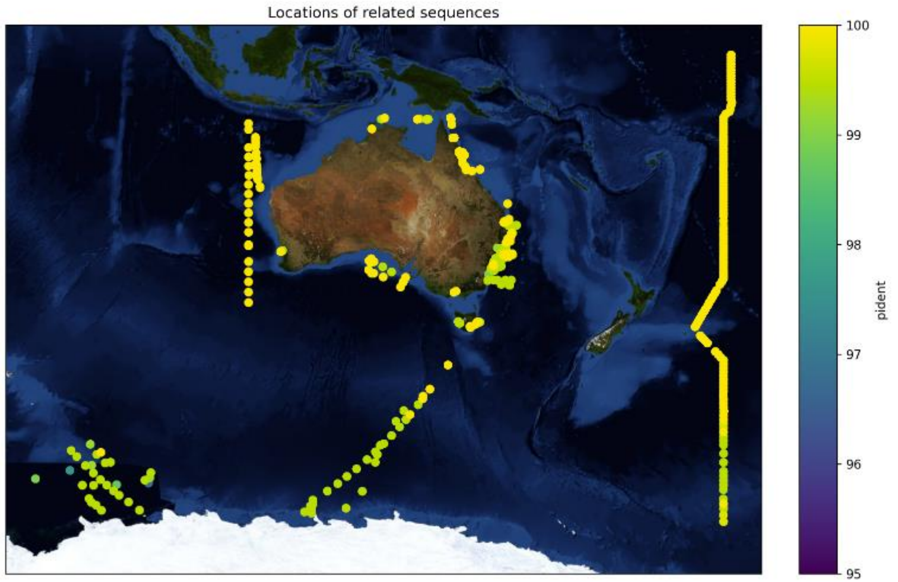
Search Parameters

qcov_hsp_perc: 60

perc_ident: 95

AGCTCCAATAGCGTATATTAAAGTTGTTGCGGTTAAAAAGCTCGTAGTTGGATTTCTGCCGAGGACGACCGGTCCGCCCTCTGGGTGAGTATCTGGCTCGGCTTGGGCATCTTCTTGGAGAAC
GTAGCTGCACTTGACTGTGTGGTGCGGTATCCAGGACTTTTACTTTGAGGAAATTAGAGTGTTTCAAGCAGGCATACGCCTTGAATACATTAGCATGGAATAATAAGATAGGACCTCGGTTCTA

BLAST search executed successfully. Click here to download the results.



Run BLAST

Filter by amplicon, taxonomy and traits

Amplicon: is

Select

Taxonomy: Select database

Domain: is

Phylum: is

Class: is

Order: is

Family: is

Genus: is

Species: is

Trait: is

Sample se

Downl

Sampl

Download metadata description

Download methods manual

l with "all/any" functions.
ldition to other contextual filters.

Interactive sample comparison

nomic Abundance Graph

Krona Plot

Amplicon <is> 18Sv4F18Sv4R_v4_eukaryote

AM Environment <is> Marine

IMOS Integrated M

Questions??

planktonr

planktonteam.github.io/planktonr

Biological Ocean Observer

shiny.csiro.au/BioOceanObserver

Contact Us

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@jaseeverett

@clairedavies

IMOS Integrated Marine Observing System

Australia's Integrated Marine Observing System is enabled by the National Collaborative Research Infrastructure Strategy (NCRIS). It is operated by a consortium of institutions as an unincorporated joint venture, with the University of Tasmania as Lead Agent.

PRINCIPAL PARTICIPANTS

University of Tasmania (Lead Agent)

SIMS is a partnership invoving four universities

ASSOCIATE PARTICIPANTS

IMOS thanks the many other organisations who partner with IMOS, providing co-investment, funding and operational support, including investment from the Tasmanian and Western Australian Governments.